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Education

- 2013–2020 Stony Brook University, Stony Brook, NY, USA
Ph.D. in Ecology & Evolution
Advisor: Dr. Jessica Gurevitch
- 2009–2013 Rochester Institute of Technology, Rochester, NY, USA
B.Sc. in Biology (*summa cum laude*), minors in French and Environmental Science

Selected Research Experience

- 2020–present **University of Konstanz**, Department of Biology
Postdoctoral Researcher
Supervisor: Dr. Mark van Kleunen
- 2018–2020 **Stony Brook University**, Department of Ecology & Evolution
Graduate Research Assistant
Supervisors: Drs. Ross Nehm and Gena Sbeglia
- 2017–2020 **Applied Biomathematics, Inc.**
Research Associate
Supervisors: Drs. Lev Ginzburg and Nicholas Friedenberg
- 2017–2018 **Brookhaven National Laboratory**, Environmental & Climate Sciences Department
Research Assistant
Supervisor: Dr. Alistair Rogers

Peer-reviewed Publications

- 2026 **Kinlock, NL**, AJS Davis, and M van Kleunen. Documenting the diversity and distributions of the World's Cultivated Plants in the WorldCuP database. *Plants, People, Planet*, ppp3.70274. [\[doi\]](#)
- 2026 Omer, A, S Dullinger, J Wessely, B Lenzner, A García-Rodríguez, A Schertler, D Moser, A Gatringer, AJS Davis, W Dawson, T Fristoe, M Grenié, **NL Kinlock**, H Kreft, J Pergl, P Pyšek, M van Kleunen, P Weigelt, M Winter, D Zurell, and F Essl. The global geography of plant invasion risk under future climate and land use changes. *Nature Ecology & Evolution*, 10(5): 952–960. [\[doi\]](#)

- 2025 **Kinlock, NL**, DW Adams, W Dawson, F Essl, J Kartesz, H Kreft, M Nishino, J Pergl, P Pyšek, P Weigelt, and M van Kleunen. Naturalization of ornamental plants in the United States depends on cultivation and historical land cover context. *Ecography*, 12: e07748. [\[doi\]](#)
- 2025 Paudel, R, TS Fristoe, **NL Kinlock**, AJS Davis, W Zhao, H Van Calster, M Chytrý, J Danihelka, G Decocq, L Ehrendorfer-Schratt, K Guo, W-Y Guo, Z Kaplan, S Pierce, J Wild, W Dawson, F Essl, H Kreft, J Pergl, P Pyšek, M Winter, P Weigelt, and M van Kleunen. Many plants naturalized as aliens abroad have also become more common within their native regions. *Nature Communications*, 16(1): 8227. [\[doi\]](#)
- 2025 Dong, B-C, R Dong, Q Yang, **NL Kinlock**, F-H Yu, and M van Kleunen. Predicting invasion success of naturalized cultivated plants in China. *Journal of Applied Ecology*, 62(3): 651–660. [\[doi\]](#)
- 2024 Guo, K, P Pyšek, M van Kleunen, **NL Kinlock**, M Lučanová, IJ Leitch, S Pierce, W Dawson, F Essl, H Kreft, B Lenzner, J Pergl, P Weigelt, and W-Y Guo. Plant invasion and naturalization are influenced by genome size, ecology and economic use globally. *Nature Communications*, 15(1): 1330. [\[doi\]](#)
- 2024 Dong, B-C, Q Yang, **NL Kinlock**, R Pouteau, P Pyšek, P Weigelt, F Yu, and M van Kleunen. Naturalization of introduced plants is driven by life-form-dependent cultivation biases. *Diversity and Distributions*, 30(1): 55–70. [\[doi\]](#)
- 2023 **Kinlock, NL**, DW Adams, and M van Kleunen. An ecological and evolutionary perspective of the historical US nursery flora. *Plants, People, Planet*, 5(1): 146–159. [\[doi\]](#)
- 2023 Fristoe, TS, J Bleilevens, **NL Kinlock**, Q Yang, Z Zhang, W Dawson, F Essl, H Kreft, J Pergl, P Pyšek, P Weigelt, JM Dufour-Dror, AN Sennikov, P Wasowicz, KB Westergaard, and M van Kleunen. Evolutionary imbalance, climate and human history jointly shape the global biogeography of alien plants. *Nature Ecology & Evolution*, 7(10): 1633–1644. [\[doi\]](#)
- 2023 Fertakos, ME, EM Beaury, NR Ford, **NL Kinlock**, DW Adams, and BA Bradley. Historical Plant Sales (HPS) database: Documenting the spatiotemporal history of plant sales in the conterminous U.S. *Ecology*, 104(7): e4106. [\[doi\]](#)
- 2022 **Kinlock, NL**, K Dehnen-Schmutz, F Essl, J Pergl, P Pyšek, H Kreft, P Weigelt, Q Yang, and M van Kleunen. Introduction history mediates naturalization and invasiveness of cultivated plants. *Global Ecology and Biogeography*, 31(6): 1104–1119. [\[doi\]](#)
- 2021 **Kinlock, NL** and SB Munch. Interaction network structure and spatial patterns influence invasiveness and invasibility in a stochastic model of plant communities. *Oikos*, 130(11): 2040–2052. [\[doi\]](#)
- 2021 **Kinlock, NL**. Uncovering structural features that underlie coexistence in an invaded woody plant community with interaction networks at multiple life stages. *Journal of Ecology*, 109(1): 384–398. [\[doi\]](#)
- 2020 **Kinlock, NL**, CM Foley, GC Sbeglia, and R Nehm. A lesson on matter and energy at the organismal scale: Linking patterns and processes across diverse taxa. *CourseSource*, 7. [\[doi\]](#)
- 2020 **Kinlock, NL**, AJ Laybourn, CE Murphy, JJ Hoover, and NA Friedenberg. Modelling

- bioenergetic and population-level impacts of invasive bigheaded carps (*Hypophthalmichthys* spp.) on native paddlefish (*Polyodon spathula*) in backwaters of the lower Mississippi River. *Freshwater Biology*, 65: 1086–1100. [doi]
- 2020 Kattge, J, G Bönisch, S Díaz, S Lavorel, IC Prentice, ..., **NL Kinlock**, ... (729 coauthors). TRY plant trait database—enhanced coverage and open access. *Global Change Biology*, 26: 119–188. [doi]
- 2019 **Kinlock, NL**. A meta-analysis of plant interaction networks reveals competitive hierarchies as well as facilitation and intransitivity. *The American Naturalist*, 194(5): 640–653. [doi]
- 2019 **Kinlock, NL**, L Prowant, EM Herstoff, CM Foley, M Akin-Fajiye, N Bender, M Umarani, HY Ryu, B Şen, and J Gurevitch. Open science and meta-analysis allow for rapid advances in ecology: A response to Menegotto et al. (2019). *Global Ecology and Biogeography*, 28(10): 1533–1534. [doi]
- 2019 Beckmann, M, K Gerstner, M Akin-Fajiye, S Ceaşu, S Kambach, **NL Kinlock**, HRP Phillips, W Verhagen, J Gurevitch, S Klotz, T Newbold, PH Verburg, M Winter, and R Seppelt. Conventional land-use intensification reduces species richness and increases production: A global meta-analysis. *Global Change Biology*, 25(6): 1941–1956. [doi]
- 2019 Peng, S, **NL Kinlock**, J Gurevitch, and S Peng. Correlation of native and exotic species richness: a global meta-analysis finds no invasion paradox across scales. *Ecology*, 100(1): e02552. [doi]
- 2018 **Kinlock, NL**, L Prowant, EM Herstoff, CM Foley, M Akin-Fajiye, N Bender, M Umarani, HY Ryu, B Şen, and J Gurevitch. Explaining global variation in the latitudinal diversity gradient: Meta-analysis confirms known patterns and uncovers new ones. *Global Ecology and Biogeography*, 27: 125–141. [doi]
- 2015 **Kinlock, NL**, BY Schindler, and J Gurevitch. Biological invasions in the context of green roofs. *Israel Journal of Ecology and Evolution*, 62(1-2): 32–43. [doi]

Scientific Presentations

- 2026 Kinlock, NL. “Cultivation as a lens to interpret plant diversity during the Anthropocene.” Research seminar, Wilfrid Laurier University, Waterloo, ON, Canada. (**Invited seminar**)
- 2024 Kinlock, NL, DW Adams, W Dawson, F Essl, J Kartesz, H Kreft, M Nishino, J Pergl, P Pyšek, P Weigelt, and M van Kleunen. “Naturalization of ornamental plants in the United States depends on cultivation and historical land cover context.” Neobiota 13th International Conference on Biological Invasions, Lisbon, Portugal. (**Oral presentation**)
- 2024 Kinlock, NL. “Remnants of cultivation and landscape history in patterns of plant invasion.” German Centre for Integrative Biodiversity Research (iDiv) seminar, Leipzig, Germany. (**Invited seminar**)
- 2022 Kinlock, NL, K Dehnen-Schmutz, F Essl, J Pergl, P Pyšek, H Kreft, P Weigelt, Q Yang, M van Kleunen. “Introduction history mediates naturalization and invasiveness of cultivated plants.” International Association for Ecology (INTECOL) 2022, Geneva, Switzerland.

(Oral presentation)

- 2022 Kinlock, NL, K Dehnen-Schmutz, F Essl, J Pergl, P Pyšek, H Kreft, P Weigelt, Q Yang, M van Kleunen. "Introduction history mediates naturalization and invasiveness of cultivated plants." Neobiota 12th International Conference on Biological Invasions, Tartu, Estonia. *(Oral presentation)*
- 2018 Kinlock, NL and J Gurevitch. "Invasive status does not dictate community-level interactions in an interaction network of woody plants." Ecological Society of America Annual Meeting, New Orleans, LA, USA. *(Poster presentation)*
- 2017 Kinlock, NL and J Gurevitch. "Experimental determination of an invaded old field plant interaction network." Early Career Researcher Symposium, Brookhaven National Laboratory, Upton, NY, USA. *(Poster presentation)*
- 2017 Kinlock, NL and J Gurevitch. "Characterizing the structure of plant interaction networks." Ecological Society of America Annual Meeting, Portland, OR, USA. *(Poster presentation)*
- 2017 Kinlock, NL. "A meta-analysis of network structure in plant-plant interaction communities." Society for Research Synthesis Methodology Annual Meeting, Montreal, QC, Canada. *(Oral presentation)*
- 2017 Kinlock, NL. "Characterizing the structure of plant interaction networks." Department of Ecology & Evolution Retreat, Stony Brook University, Stony Brook, NY. *(Oral presentation)*
- 2013 Kinlock, NL and AC Tyler. "Does the history of small, urban and suburban wetlands influence the biodiversity of aquatic invertebrate communities?" College of Science Seminar, Rochester Institute of Technology, Rochester, NY. *(Oral presentation)*
- 2012 Kinlock, NL and AC Tyler. "Analyzing the effectiveness and efficiency of different methods for invasive *Typha* spp. removal in created wetlands." Summer Undergraduate Research Symposium, Rochester Institute of Technology, Rochester, NY. *(Oral presentation)*
- 2011 Kinlock, NL, S Loftus, N Marshall, J Skufca, and M Twiss. "Comparison of plankton dynamics in nearshore and main channel areas of the St. Lawrence River." Summer Symposium on Undergraduate Research Experiences, Clarkson University, Potsdam, NY. *(Oral presentation)*

Supervision and Mentorship

- 2025 Yu Hamada* – Master's thesis: Intraspecific diversity as a proxy of intraspecific genetic and phenotypic variation and potential for naturalization success – University of Konstanz
- 2025 Valentina Zickenberg* – Master's thesis: The influence of historical land-use on the naturalization of cultivated plants in Germany – University of Konstanz
- 2024 Jonathan Mo* – Bachelor's thesis: Tracing signatures of imperial possession in cultivated plants of colonial botanic gardens and the naturalized plants of today – University of Konstanz
- 2017 Maria Zambuto – Undergraduate research project: Comparing intra- and interspecific com-

petition of spotted knapweed and orchardgrass – Stony Brook University

* *Co-supervised students with Mark van Kleunen*

Teaching Activities

Instructor

- 2024–present Global Change Ecology and Plants – Masters-level course with lecture, workshops, and student-designed projects at the University of Konstanz
- 2023–present Plant Physiology – practical course for advanced undergraduate students at the University of Konstanz
- 2019 Biometry – statistics course for Masters and Ph.D. students at Stony Brook University

Teaching Assistant or Guest Lecturer

- 2018 Plant Ecology* – guest lecturer for one month
- 2018 Ecology*
- 2016–2017 Biometry†
- 2015 Landscape Ecology Laboratory*
- 2015 Plant Diversity*
- 2014–2019 Fundamentals of Biology: Organisms to Ecosystems*
- 2014 Applied Ecology and Conservation Biology Laboratory*
- 2013 Fundamentals of Scientific Inquiry in the Biological Sciences I*
- * *Courses for Bachelors students at Stony Brook University*
- † *Course for Masters and Ph.D. students at Stony Brook University*

Recognition

- 2022 Shortlisted for the Harper Prize, *Journal of Ecology* annual award for the best paper by an early career author
- 2020 Best Teaching Assistant Award, Department of Ecology & Evolution, Stony Brook University
- 2015, 2017 Lawrence B. Slobodkin Award for Research in Ecology (**\$750, \$550**)
- 2014 Tinker Foundation Field Research Grant (**\$1,997**)
- 2014 Departmental Excellence Research Award (**\$1,000**)
- 2013 Graduate Council Fellowship, Stony Brook University (**\$125,000**) *Awarded annually to ten incoming doctoral students at Stony Brook University.*
- 2013 John Wiley Jones Award for Outstanding Students in Science (**\$250**)

Professional Services

Reviewer for *Biological Invasions; Diversity and Distributions; Ecography; Ecology; Ecology and Evolution; Ecology Letters; Ecosphere; Functional Ecology; Frontiers of Ecology & Evolution; Global Ecology and Biogeography; Israel Journal of Ecology and Evolution; Journal of Biogeography; Journal of Ecology; Journal of Experimental Botany; Nature Communications; Nature Ecology & Evolution; Nature Plants; Plants, People, Planet; PLOS ONE; Proceedings B; Science; and Scientific Reports.*

Member of the Ecological Society of America and the American Society of Naturalists

Professional Development

- 2024 InvaPact II workshop participant, Université Paris-Saclay
Workshop about identifying, quantifying, and communicating invasive species impacts
- 2021–present Global Naturalized Alien Flora (GloNAF) workshop participant, GloNAF Consortium
Annual consortium meeting to update research progress, identify gaps, and plan projects
- 2019 American Association for the Advancement of Science (AAAS) Project 2061 workshop participant, Brookhaven National Laboratory
Workshop introducing curriculum materials designed for biological science educators to help students meet the Next Generation Science Standards
- 2016–2019 Introduction to R workshop organizer, Stony Brook University
Workshop introducing first-year Masters and Ph.D. students to the R programming language
- 2014 Tropical Plant Systematics course participant, Organization for Tropical Studies, Costa Rica
Five-week graduate course on the phylogeny and identification of neotropical plants with a research component
- 2013–2014 Improvisation for Scientists workshop participant, Alan Alda Center for Communicating Science, Stony Brook University
Weekly workshop to improve communication skills, tailored to scientists
- 2012 Undergraduate stewardship liaison in the Great Lakes Innovative Stewardship Through Education Network, Rochester Institute of Technology
Position designing and implementing wetland mitigation, invasive species management, and community outreach projects

Public Outreach

- 2022 **Blog feature.** “Plants Sold – Increasing % of Exotics.” Interviewed by Faith Campbell for the Center for Invasive Species Prevention. [\[html\]](#)
- 2019 **Public talk.** “Invasive plants in the northeast U.S.: Combining natural history and ecology.” Friends of the Long Pond Greenbelt Sunday Talks Series, Bridgehampton, NY, USA.

2017 **Public talk.** “Community structure of an invaded old field plant interaction network.” Yale-Myers Summer Seminar Series, Eastford, CT, USA.

Skills

Programming languages R, Python, Julia, C++, Bash

High-performance cluster computing

Git ([GitHub/nlkinlock](#))